Year 2 Assessed Problems

Semester 1

Assessed Problems 1

SOLUTIONS TO BE SUBMITTED
ON CANVAS BY

Wednesday 15th October
at 17:00

Quantum Mechanics 2 - Problem Sheet 1

A particle of mass m is confined to the region $0 < x < d$. Within this region it is described by the wavefunction:

$$\Psi(x,t) = A\sin(2\pi x/d)\, e^{-iEt/\hbar}$$

while outside this region the wavefunction $= 0$.

Calculate the value of the normalisation constant A that gives an integrated probability of 1 for finding this particle within the region $0 < x < d$. Assume that A is a real number. [4]

Calculate the expectation value of the kinetic energy for this particle. [2]

Calculate the expectation value $<x>$ for the position of this particle.
What is the value of the wavefunction at position $x = <x>$?
Comment on your answers. [4]

1 Assessed – Cycles

For this problem you may want to refer to your Year 1 Temperature and Matter course.

We consider an *ideal gas*, which follows the equations of state $pV = n\mathrm{R}T$ and $U = \frac{3}{2}N\mathrm{k}_\mathrm{B}T$, where the symbols have their usual meaning. We recall that the work done on the gas in a reversible transformation is $-\int p\,\mathrm{d}V$ and the heat capacity C_V is a *constant* such that $đQ = C_V\,\mathrm{d}T$ in a transformation that keeps the volume constant.

1. Give an expression of the work W done *on* the system and the heat Q transferred *to* the system in a isochoric transformation which changes the temperature from T_1 to T_2. **[2]**
2. Give an expression of the work W done *on* the system and the heat Q transferred *to* the system in an isothermal reversible expansion which changes the volume from V_1 to V_2. **[2]**

We now consider a cycle consisting of two isothermal transformations at temperatures T_1 and T_2 ($T_2 > T_1$) and two isochoric transformations at volumes V_1 and V_2 ($V_2 > V_1$).

3. Draw the cycle on a pV diagram. Indicate the direction of the cycle for it to produce net work. For each of the four segments of the cycle, indicate the sign of the work done *by* the system and the heat transferred *to* the system. **[2]**
4. Draw the cycle on a VT diagram (T on the abscissa). Indicate the direction of the cycle for it to produce net work. **[2]**
5. You have 2 heat sources with temperatures T_1 and T_2. Can you operate the cycle above on an ideal gas in a reversible fashion? Explain why. **[2]**